Open Source for Regulated Manufacturing

GMP-Compliant IIoT Data Logging with Apache IoTDB

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The moderne Pharmaceutical Manufacturing

Why Data Integrity Matters

- Regulated Industries: Crucial for life sciences, including Pharma, MedTech, and Biotech manufacturing.
- The Core Conflict: High-frequency IIoT industrial sensor data velocity colliding directly with strict compliance demands.
- The Shift: Evolving global frameworks require absolute transparency across the entire data lifecycle.
- Motivation: Can open source serve as a compliant historian backbone and pass a rigorous FDA/EU GMP inspection

The 2026 Regulatory Storm

- The Hook: Big Pharma is facing a massive regulatory shift heading into 2026.
- Annex 11 Overhaul: Expanding from 1,500 to 10,000 words, bringing highly detailed requirements for data integrity, cloud computing, and IT security.
- The New Annex 22: Created specifically to regulate AI-supported and automated systems within GMP environments.
- The Collision: High-frequency IIoT devices are generating petabytes of data meeting the CPU/AI compliance era.

The Regulatory Villain

- Traditional Historians: Proprietary ecosystems charging extortionate fees per data tag.
- The Scaling Break: Too expensive to scale; when handling high-frequency IIoT velocity, they break or bankrupt you.
- The Burning Question: Can open-source software actually pass a rigorous FDA/EU GMP inspection better than proprietary options?

Graphic depicting AI/LLM restrictions and the introduction of Annex 22 in 2026

Compliance in 60 Seconds: FDA vs. EU (Part 1)

- 🇺🇸 FDA 21 CFR Part 11 (US Regulation)
  - Defines rules for electronic records & electronic signatures.
  - Ensures electronic systems are trustworthy, secure, and legally equivalent to paper.
  - Key Question: **"Are electronic records & signatures trustworthy?"**

Compliance in 60 Seconds: FDA vs. EU (Part 2)

- 🇪🇺 EU Annex 11 (GMP Guideline)
  - Governs computerized systems in GMP environments.
  - Covers the full system lifecycle including validation, risk management, and supplier governance.
  - Key Question: **"Is the entire system validated, controlled, and compliant?"**

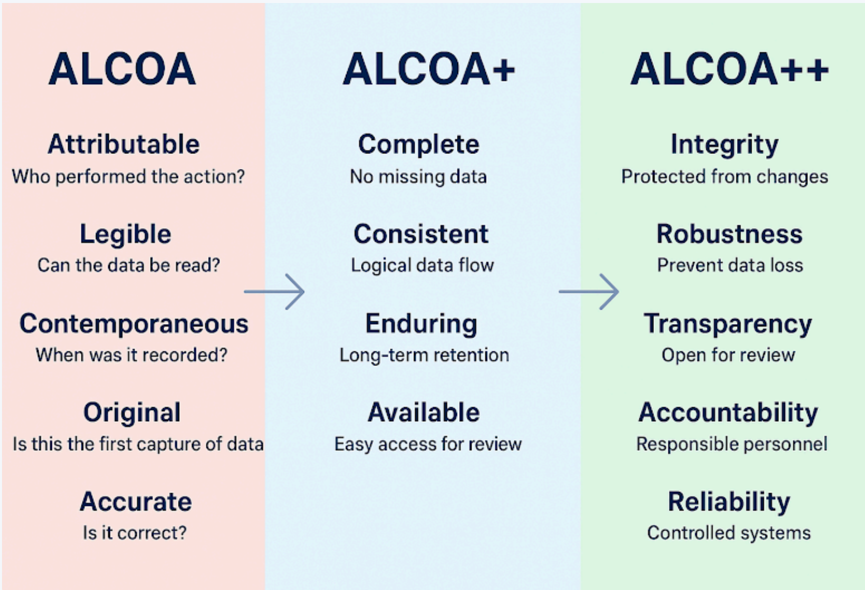
Compliance in 60 Seconds: The Takeaway

- Common Goal: Ensure data integrity, traceability, and compliance.
- The Digital Shift: Enable audit-ready, reliable digital processes instead of paper.
- Takeaway: Both regulations ensure digital GMP data is trustworthy across the full data lifecycle.

GMP Requirements for IIoT Data

Demystifying ALCOA++ (The Core Framework)

- The Gold Standard: ALCOA+ is a framework ensuring the accuracy, consistency, and reliability of data in highly regulated industries.
- The Evolution: Original ALCOA metrics expand into ALCOA+ and ALCOA++ to safely cover automated, high-velocity infrastructure.



ALCOA: The Core 5 Principles

Principle	Definition
Attributable	Data must be clearly linked to the person or device that generated it
Legible	Data must be readable, permanent, and accessible throughout its lifecycle
Contemporaneous	Data must be recorded at the time of the activity, not retrospectively
Original	Data must be the first capture of information, not a copy or transcription

ALCOA+: The 4 Extensions

Principle	Definition
Complete	All data, including deletions or modifications, must be retained and traceable
Consistent	Data must be reliable, reproducible, and aligned with expected patterns
Enduring	Data must be stored on durable media and remain accessible for required retention
Available	Data must be readily accessible to authorized personnel when needed

ALCOA++: The 5 Modern IIoT Safeguards

The Gap: Standard IT open-source tools fail ALCOA++ out of the box because mutable databases allow altering logs without audit trails.

The 5 Security & Telemetry Dimensions:

Principle	Definition
Integrity	Protected from unauthorized changes
Robustness	Prevent data loss at high velocity
Transparency	Open infrastructure design and review logic
Accountability	Explicitly traceable to responsible personnel

Main Elements of 21 CFR Part 11

- Validation: Systems must be validated to perform exactly as intended
- Security: Access to electronic records restricted to authorized users
- Record Retention: Records must be securely retained in an unalterable environment
- Audit Trails: Every action captured automatically with precise date/time stamps
- Electronic Signatures: Unique and indissolubly linked to their respective records
- Training: Personnel must be trained and accountable under Part 11 requirements

The Core Challenge

Why Traditional Historians Struggle (Commercial Constraints)

- Proprietary Licensing: Restrictive models based on strict data tag counts or ingestion volumes
- Vendor Lock-In: Limited architecture flexibility and high vendor dependency
- Expensive Scaling: Extreme costs to scale physical storage, high-availability, and retention
- Validation Effort: Exceptionally high validation and customization efforts inside a GAMP 5 framework

Why Traditional Historians Struggle (Technical Constraints)

- Siloed Architectures: Rigid data models offering minimal metadata, data lineage tracking, and query reproducibility
- Velocity Bottlenecks: Inability to handle high-frequency sensor and rapid PLC time-series streams natively
- Weak Auditing: Demands intense layout customization to comply with automated ALCOA+ principles
- Integration Gaps: Poor interface performance when talking to modern streaming analytics Engines

The Industrial Data Reality: High-frequency sensor & PLC data needs rapid ingestion, massive capacity, and strict ALCOA+ audit trails

The Role of Open Source

Open Source in Regulated Environments

- The Transparency Advantage: Closed-source platforms act as compliance "black boxes". Open source allows auditors to inspect data processing logic directly.
- Interoperability: Open APIs and standardized file formats eliminate downstream analytical silos
- Ecosystem Trust: The Apache Software Foundation (ASF) provides community governance that minimizes vendor lock-in risks

Open Infrastructure Data Commons (OIDC)

- Real-World Baseline: An open-source software platform leveraging Apache technologies for real-time monitoring and data preservation.
- UN Recognition: Formally published within the United Nations STI Solutions Book, proving open infrastructure handles mission-critical, long-term data preservation.

Screenshot of page 36 highlighting Open Infrastructure Data Commons using iotdb.apache.org

The Apache Solution

Apache IoTDB as a GMP Compliant IIoT Historian

- Purpose-Built: Optimized directly for high-frequency industrial sensors and PLC time-series streams
- Core Compliance features: Structural mechanisms for unbending audit trail tracking, data versioning, and lineage
- Security Controls: Built-in user authentication, role-based access control (RBAC), encryption, and automated data retention management
- UN Recognition: Selected as one of the **Top 60 Global Innovation Cases** by the United Nations STI Forum

Core Enabler: Apache TSFile

- Immutable Design Pattern: A columnar file layout that is append-only, serving as a regulatory dream because data cannot be silently modified or manipulated
- Data Segmentation: Isolated physical data layout blocks enable clean validation boundaries and targeted spot-checks
- Ecosystem Hybridity: Combining TSFile with Apache Iceberg catalogs allows complex systems to track full historical query lineage and reproduce query states

Apache Projects for an End-to-End Pipeline

- Data Integration & Edge: Apache PLC4X parses OPC-UA/Modbus protocols natively. Apache NiFi handles automated data flow and integration.
- Messaging & Ingestion: Apache BifroMQ serves as a high-throughput MQTT broker, feeding distributed Apache Kafka data streams.
- Stream Processing & Rules: Apache Flink or Apache Beam execute real-time compliance filtering and threshold validation.
- Storage & Analytics: Apache IoTDB (hot storage) tracks time-series, while Apache Iceberg handles cold archives, analyzed via Apache Superset.

Reference Architecture Map

End-to-end component diagram highlighting data paths from Edge Protocols through BifroMQ/Kafka up to IoTDB/TSFile and Iceberg Data Lakes

Compliance and Validation

Compliance Considerations

- Annex 11 Mapping: Matching specific clauses (e.g., electronic audit trails, system security, automated printouts) to configuration profiles in Apache IoTDB
- 21 CFR Part 11 Mapping: Utilizing integrated user authorization matrices, electronic signatures concepts, and unalterable time-stamped logs
- Validation Boundaries: Clearly defining software boundaries between underlying operating systems, infrastructure, and containerized Apache applications

Computer System Validation with GAMP 5

- The 9 Steps of Annex 11 Computer System Validation (CSV):

1. Risk Assessment
2. Supplier Qualification
3. Requirements (URS)
4. Design & Specifications
5. IQ/OQ/PQ Testing
6. Data Integrity Controls
7. SOPs & Training
8. Change Control & Review
9. Operation & Monitoring

The 3 -Tier Testing Blueprint (Step 5)

- Installation Qualification (IQ): Repeatable deployment checking using automated, reproducible container setups
- Operational Qualification (OQ): Verifying database failover operations, security profiles, user access, and automated data retention policies
- Performance Qualification (PQ): Simulating maximum industrial high-frequency IIoT velocity to guarantee stability and zero data loss under peak load

Conclusion

Key Takeaways

- The Open Advantage: Open-source ecosystems fully support transparent, scalable, and compliant industrial data platforms
- Future-Proof Storage: Combining Apache IoTDB, TSFile, and Iceberg builds a modern, cost-effective, and audit-ready data lake landscape
- Strategic Shift: Adopting open source enables factories to navigate evolving regulations without getting trapped by legacy per-tag vendor licensing fees

Q&A

- Thank you for attending!
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